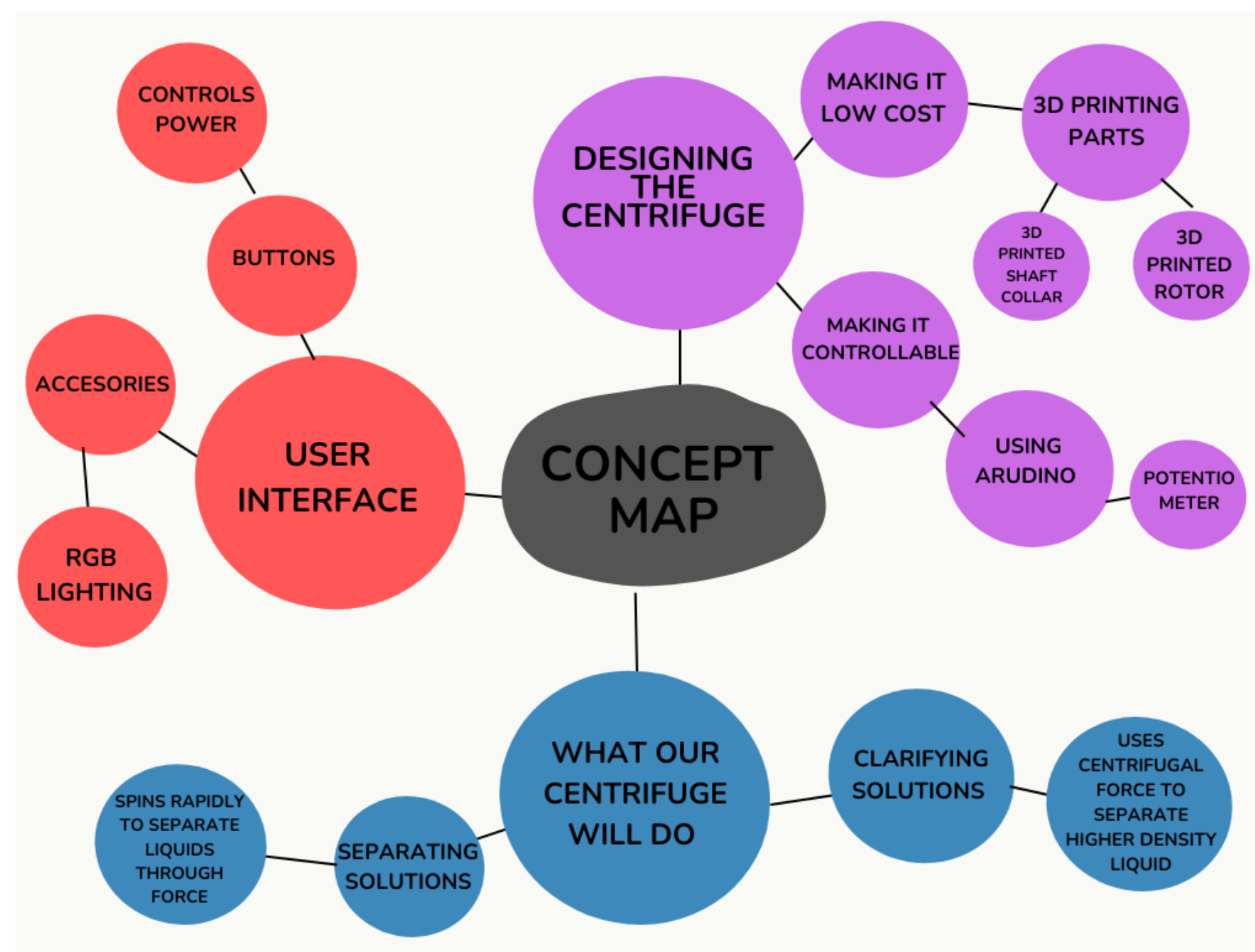


Problem Statement

Every day whether it be in academia, research, or industry jobs, the separation of substances and liquids is crucial to conducting tests, analyzing data, and getting a certain substance out of a liquid. Performing these processes can cost thousands of dollars and it can take lots of time to conduct tests due to their slow rate of movement or having to adjust for fixed speeds.

Concept Map



Our Solution

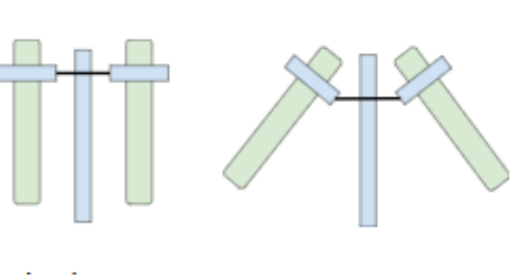
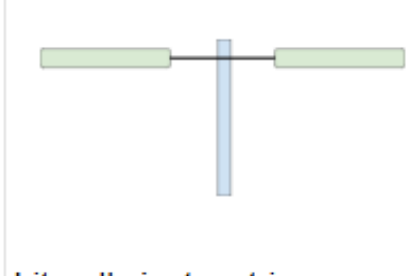
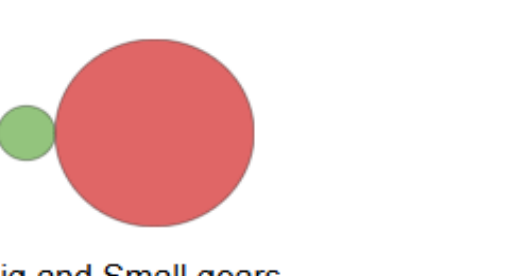
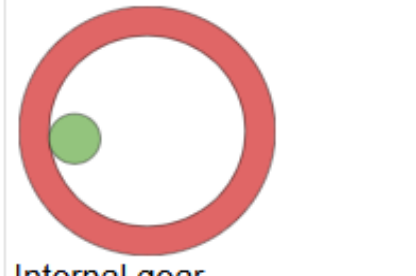
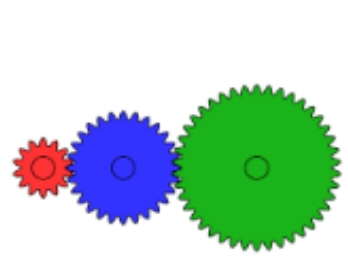
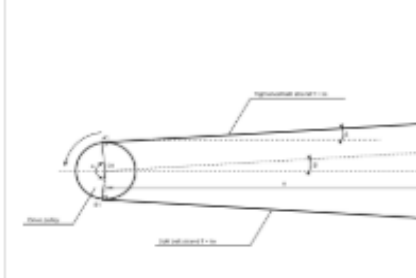
- Cost Effectiveness: We used 3d printed material to reduce costs
- Portability: We used reduced thickness in the parts in order to allow for a light, east to transport product
- Controllability: We used Arduino to control the speed/RPM of our centrifuge



Our Team

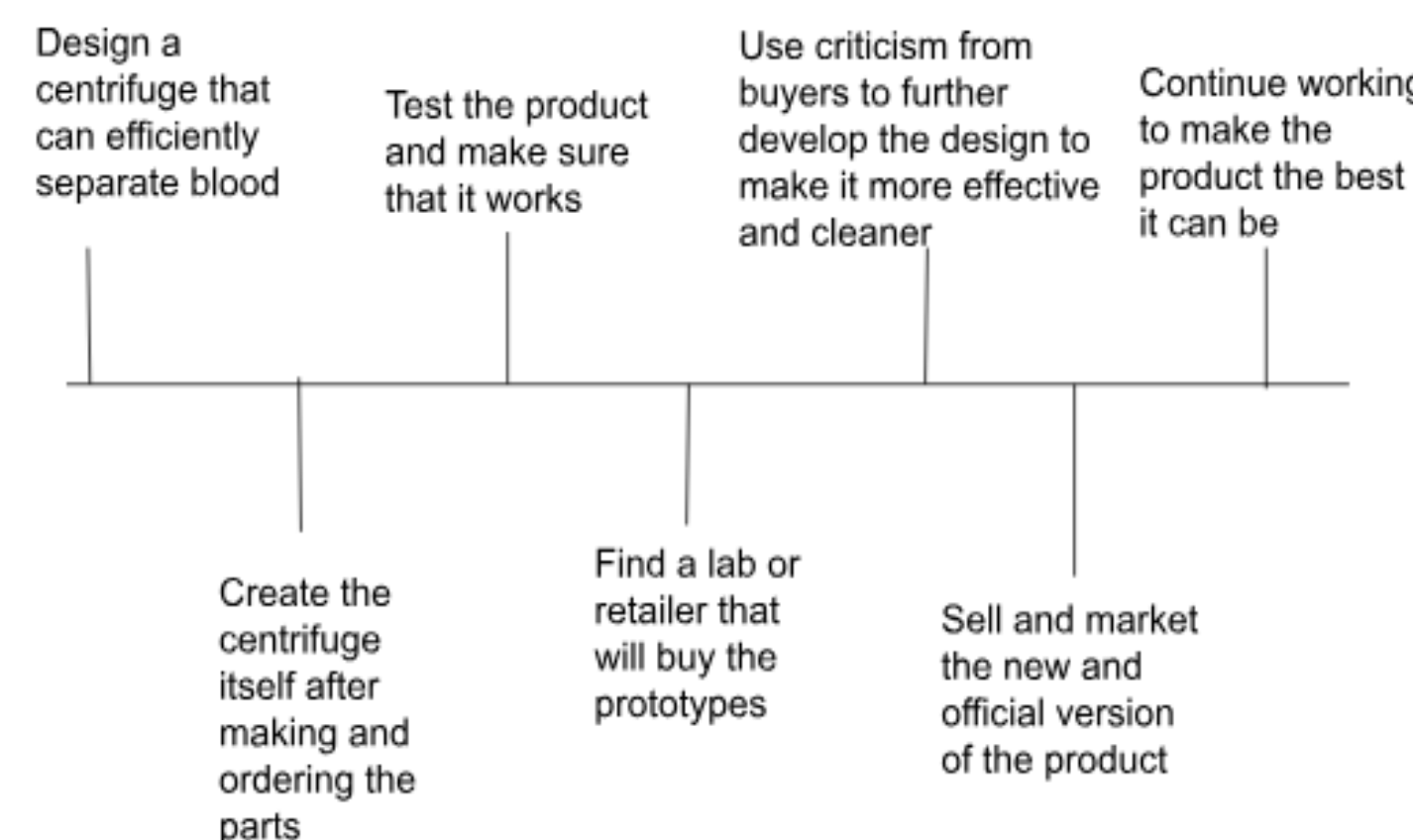
- R&D Engineer: **Donovan Zoma** – Designed CAD/NX parts, gathered materials
- R&D Engineer: **Sami Miah** – Helped create our poster, wrote the programs for our Arduino model
- Quality Engineer and/or Process Engineer: **Adam Ong** – Helped review part designs, created the morph matrix, created the hardware, managed parts brought in,
- Product Manager: **Vaibhav Singotam** – Helped manage deadlines, assigned tasks effectively, created the concept map, and worked with team members to assign tasks based on teammates desired roles and skills

Morph Matrix

	Option 1	Option 2	Option 3	Option 4
Vial housing	 Solid vial housing	 Swinging	 Literally just a string	
Gearing - Red is directly powered, green is driving centrifuge	 Planetary gearing	 Big and Small gears	 Internal gear	
Connection - How is everything powered. Note: The gearing row was made with mostly gears in mind. Some options in this row may not work as shown, but may be able to be modified so that they do work in a similar fashion	 Gears	 Chain & Sprocket	 String/Belt & Pulleys	
				Directly driven

Cost Analysis & Timeline

- The total cost of our Centrifuge is 75\$/8 Hours
- The future costs of our product is roughly between 21,120 and 21,160 \$ / 150 Hours



References

- Anne Marie Helmenstine, *What centrifugation is and why it's used*
- Encyclopedia Britannica, *Centrifuge*
- *HOW TO MAKE A LAB CENTRIFUGE*, [Blog]

Acknowledgements

Thank you to Dr. Harris and the TA's for helping us through times when we did not know how to use new technology like the Arduino which none of us had previous experience with and for providing funding for our project.